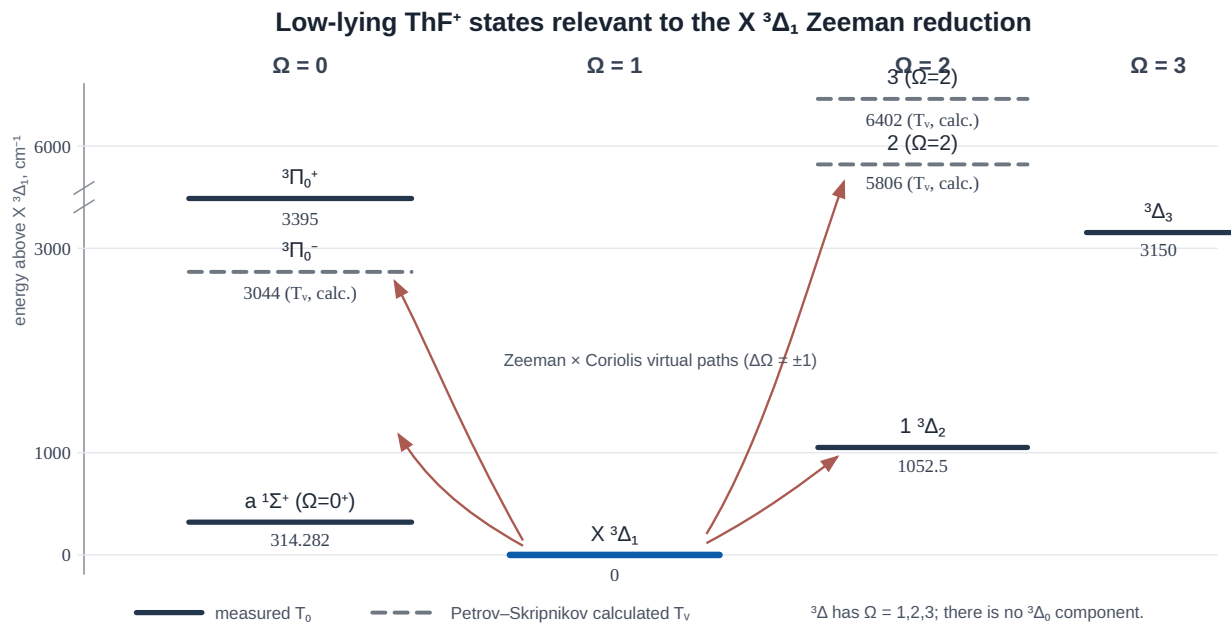


Effective Zeeman Hamiltonian for $^{232}\text{ThF}^+ X^3\Delta_1$

Second-order Brown–Carrington/Van Vleck reduction, effective $\mathbf{G}$ tensor, and comparison with $\text{ThF}^+/\text{ThO}/\text{HfF}^+$ theory and spectroscopy.

Objective. Replace the explicit nearby $\Omega = 0^\pm$ and $\Omega = 2$ electronic states by an effective Hamiltonian acting only in the $X^3\Delta_1$ rovibronic manifold. To second order in off-diagonal electronic–rotational couplings and first order in the magnetic field $\mathbf{B}$, the response can be represented by one rovibronic effective Zeeman tensor $\mathbf{G}_X^{\text{eff}}$ plus the nuclear Zeeman term. A separate leading scalar rotational g_R is then redundant unless it multiplies an operator linearly independent of $\mathbf{B}\cdot\mathbf{G}\cdot\mathbf{J}$.



Low-lying electronic structure relevant to the reduction. Solid levels are measured T_0 values; dashed levels are calculated vertical energies T_v used in the 2025 Petrov–Skripnikov basis. Leading Zeeman–Coriolis paths from $X^3\Delta_1$ connect to $\Omega = 0$ and $\Omega = 2$ states. A $^3\Delta$ term has $\Omega = 1, 2, 3$ —there is no $^3\Delta_0$ component; the relevant $\Omega = 0$ perturbors are $a^1\Sigma^+$ and $^3\Pi_{0\pm}$. Energies from Refs. [1,5].

1. Notation and model space

Symbol	Meaning
$X^3\Delta_1$	eEDM-sensitive electronic ground state of $^{232}\text{ThF}^+$; its energy defines zero in the diagram.
Ω	Projection of electronic angular momentum $\mathbf{J}^e$ on the internuclear axis $\hat{n}$.
$\mathbf{J}; J$	Molecular angular momentum excluding nuclear spin; J is its quantum number.
$\mathbf{I}_F; \mathbf{F}; M_F$	^{19}F nuclear spin ($I_F = 1/2$), $\mathbf{F} = \mathbf{J} + \mathbf{I}_F$, and laboratory-axis projection of $\mathbf{F}$.
$P; Q = 1 - P$	Projectors onto the retained $X^3\Delta_1$ rovibronic model space and the eliminated electronic space.
$H_0; E_X; R_X$	Unperturbed field-free Hamiltonian, model-space energy, and reduced resolvent $R_X = Q(E_X - QH_0Q)^{-1}Q$.
$\mu_B; \mu_N$	Bohr and nuclear magnetons.
$\mathbf{L}^e, \mathbf{S}^e, \mathbf{J}^e$	Electronic orbital, spin, and total angular momenta; $\mathbf{J}^e = \mathbf{L}^e + \mathbf{S}^e$.
B_{rot}	Rotational constant; Ref. [1] uses 0.243 cm^{-1} .
g_S	Free-electron spin g factor in the Petrov sign convention, $g_S = -2.0023$.
e/f	Opposite-parity members of the Ω doublet.

2. Brown–Carrington / Van Vleck reduction to second order

Write $H = H_0 + V$, with V containing the off-diagonal rotation–electronic and Zeeman interactions. The second-order model-space contribution is schematically

$$H_{\text{eff}}^{(2)} = PVR_XVP, \quad R_X = Q(E_X - QH_0Q)^{-1}Q, \quad (1)$$

with the standard Hermitian symmetrization understood when different vertices occur. The vertices relevant to the term linear in $\mathbf{B}$ are

$$V_Z = \mu_B(\mathbf{L}^e - g_S\mathbf{S}^e) \cdot \mathbf{B}, \quad (2)$$

$$V_C = -2B_{\text{rot}}\mathbf{J} \cdot \mathbf{J}^e, \quad (3)$$

so that

$$H_{Z,\text{eff}}^{(2)} = PV_ZR_XV_CP + PV_CR_XV_ZP. \quad (4)$$

V_Z is the electronic Zeeman operator and V_C is the Coriolis (rotation–electronic or L -uncoupling) interaction. This is the same mechanism used in the heavy $^3\Delta_1$ calculations for ThO and HfF $^+$: the small body-frame magnetic moment is renormalized by virtual mixing with nearby $\Omega = 0^\pm$ and $\Omega = 2$ states [6,7].

3. Minimal effective Zeeman tensor inside $|\Omega| = 1$

Define the *complete rovibronic* response, not a bare fixed-nuclei electronic tensor, by

$$H_Z^X = \mu_B\mathbf{B} \cdot \mathbf{G}_X^{\text{eff}} \cdot \mathbf{J} - g_{I,F}\mu_N\mathbf{B} \cdot \mathbf{I}_F + H_Z^{(>2)}. \quad (5)$$

Here $g_{I,F}$ is the fluorine nuclear g factor. $H_Z^{(>2)}$ denotes higher-order *effective-Hamiltonian* corrections beyond the second-order electronic/rotational reduction; it does not denote a term quadratic in B .

In one convenient real representation of the Ω doublet,

$$\mathbf{G}_{e/f}^{\text{eff}} = \begin{pmatrix} G_\perp \pm G_\Delta & 0 & 0 \\ 0 & G_\perp \mp G_\Delta & 0 \\ 0 & 0 & G_\parallel \end{pmatrix}. \quad (6)$$

The coefficients are

$$G_\parallel = G_{zz}, \quad G_\perp = \frac{G_{xx} + G_{yy}}{2}, \quad G_\Delta = \frac{G_{xx} - G_{yy}}{2}. \quad (7)$$

Thus $G_\parallel$ is the body-axis response, $G_\perp$ the parity-even transverse response, and G_Δ the parity-dependent transverse anisotropy. The upper/lower sign depends on the e/f phases and choice of transverse body axes. In irreducible spherical-tensor form,

$$G_0^{(2)} = \frac{2G_{zz} - G_{xx} - G_{yy}}{\sqrt{6}} = \sqrt{\frac{2}{3}}(G_\parallel - G_\perp), \quad G_{\pm 2}^{(2)} \propto G_\Delta, \quad (8)$$

where $q = 0, \pm 2$ labels the molecule-fixed rank-2 spherical components. The $q = \pm 2$ piece is the effective-tensor analogue of the primed parity-dependent Zeeman operators in Brown–Carrington-style $^3\Delta_1$ Hamiltonians. Zhou, Island, and Grau give the explicit spherical-tensor/ $3j$ – $6j$ machinery for the $|J, \Omega, I_F, F, M_F\rangle$ basis; their Appendix B can be extended from the axial term to these $q = 0, \pm 2$ pieces [4].

4. Exact second-order mapping to the Petrov form

At zero electric field, the heavy- $^3\Delta_1$ second-order result can be written [6]

$$g_{e/f}(J) = -\frac{G_0}{J(J+1)} + C_{0^\pm} + C_2 \frac{(J+2)(J-1)}{2J(J+1)}. \quad (9)$$

G_0 is the diagonal body-axis electronic Zeeman parameter before the nonadiabatic second-order correction. The electronic-state sums are

$$C_{0^\pm} = \sum_{n:\Omega_n=0^\pm} \frac{G_{\perp n}\Delta_n}{E_X - E_n}, \quad C_2 = \sum_{n:\Omega_n=2} \frac{G_{\perp n}\Delta_n}{E_X - E_n}. \quad (10)$$

Here n labels an eliminated state of energy E_n ; $G_{\perp n} = \langle X|L_\pm^e - g_S S_\pm^e|n\rangle$ (or its Hermitian conjugate) is the transverse magnetic matrix element, and $\Delta_n = 2B_{\text{rot}}\langle n|J_\pm^e|X\rangle$ is the Coriolis matrix element. Only the phase-invariant product $G_{\perp n}\Delta_n$ enters.

Since

$$\frac{(J+2)(J-1)}{2J(J+1)} = \frac{1}{2} - \frac{1}{J(J+1)}, \quad (11)$$

defining $\bar{C}_0 = (C_{0^+} + C_{0^-})/2$ gives the spectrum of a J -independent tensor *exactly through second order*:

$$g_{e/f}(J) = -G_\perp - \frac{G_\parallel - G_\perp}{J(J+1)} \pm G_\Delta \quad (12)$$

with

$$G_\perp = -\left(\bar{C}_0 + \frac{C_2}{2}\right), \quad G_\parallel = G_0 - \bar{C}_0 + \frac{C_2}{2}, \quad G_\Delta = \frac{C_{0^+} - C_{0^-}}{2}. \quad (13)$$

This is the direct bridge from the Brown–Carrington effective-Hamiltonian expansion to the *ab initio* transition matrix elements.

5. ThF⁺ coefficients from the 2025 seven-state model

Petrov and Skripnikov include $X^3\Delta_1$, $a^1\Sigma^+$, $^3\Pi_{0\pm}$, and three $\Omega = 2$ states. Their $X \leftrightarrow a$ magnetic and Coriolis matrix elements are each increased by 8.8% to reproduce the measured 5.29 MHz Ω -doubling, and they use $G_0 = 0.047$ to reproduce the measured $J = 1$, $F = 3/2$ magnetic response [1–3].

Second-order sum	Value	Effective coefficient	Value
$C_{0+} (a^1\Sigma^+ + ^3\Pi_{0+})$	-6.175×10^{-4}	$G_{\parallel}$	0.04680
$C_{0-} (^3\Pi_{0-})$	-2.063×10^{-4}	$G_{\perp}$	1.028×10^{-3}
C_2 (three $\Omega = 2$ states)	-1.232×10^{-3}	G_{Δ}	-2.06×10^{-4}

A representative real-axis tensor is

$$\mathbf{G}_X^{\text{eff}} \simeq \text{diag} \left(8.22 \times 10^{-4}, 1.233 \times 10^{-3}, 4.680 \times 10^{-2} \right), \quad (14)$$

with $x \leftrightarrow y$ and the sign of G_{Δ} convention dependent. The invariant axial rank-2 amplitude is $G_0^{(2)} \simeq 3.74 \times 10^{-2}$.

J	1	2	3	4	5
$\bar{g}_J$	-2.391×10^{-2}	-8.656×10^{-3}	-4.842×10^{-3}	-3.316×10^{-3}	-2.553×10^{-3}

These are rovibronic J -level values before hyperfine/nuclear-Zeeman recoupling. For $J = 1$, $I_F = 1/2$, $F = 3/2$, the $\mathbf{J}$ contribution projects onto $\mathbf{F}$ with factor $2/3$.

6. Checks in heavy $^3\Delta_1$ systems and practical conclusion

- **ThF⁺**. Ng *et al.* measured $|g_{F=3/2}| = 0.0149(3)$ and $|\Delta g| = 3(3) \times 10^{-4}$ [2]. Petrov–Skripnikov obtain $\Delta g = 2.3 \times 10^{-4}$ from the full seven-state calculation [1]. The second-order tensor gives about 2.7×10^{-4} after $J \rightarrow F$ projection, leaving a useful ~ 15 – 20% benchmark residual.
- **ThO**. Measured multi- J g factors depart strongly from pure $1/[J(J+1)]$ scaling, and Petrov *et al.* reproduce this with the same $\Omega = 0, 2$ mixing mechanism [6].
- **HfF⁺**. Petrov–Skripnikov–Titov obtain $G_{\parallel} = 0.0115$ *ab initio* versus 0.0118 from an experimental fit and model the doublet-dependent response [7]. Meyer and later JILA treatments provide the Brown–Carrington primed/unprimed operator language [8,9].
- **Implementation**. Ng’s thesis gives the polarized-state differential- g behavior, while Zhou–Island–Grau Appendix B provides an explicit spherical-tensor Hamiltonian suitable for extension to the full $\mathbf{G}$ tensor [3,4].

Working conclusion for $J = 1$ – 5 . Use one constant three-parameter tensor ($G_{\parallel}, G_{\perp}, G_{\Delta}$) as the complete leading rovibronic Zeeman interaction, plus the fluorine nuclear Zeeman term. Do *not* add an independent leading g_R unless it multiplies an operator linearly independent of $\mathbf{B} \cdot \mathbf{G} \cdot \mathbf{J}$. Benchmark the tensor against exact diagonalization of the published seven-state ThF⁺ Hamiltonian for $J = 1, \dots, 5$; only the residual J dependence should motivate additional Brown–Carrington Zeeman-distortion operators.

Next order / *ab initio* requirements

At third order and linear order in $\mathbf{B}$, generic Van Vleck terms contain one Zeeman vertex and two additional off-diagonal vertices, e.g.

$$PV_Z R_X V_C R_X V_C P + \text{permutations} - \text{folded/normalization terms}. \quad (15)$$

These generate J -dependent renormalizations of $G_{\parallel}$, $G_{\perp}$, and G_{Δ} . Additional inputs are (i) excited-state-to-excited-state Coriolis matrix elements $\langle n | J_{\pm}^e | m \rangle$, (ii) excited-state-to-excited-state and diagonal magnetic matrix elements $\langle n | L^e - g_S S^e | m \rangle$, (iii) accurate electronic energies, and, only if intrinsically F -dependent corrections are desired, (iv) off-diagonal hyperfine matrix elements. A useful first test is to retain J -dependent rotational energies in the denominators as a partial resummation before fitting new effective constants.

Sign convention. Petrov defines $E_Z = -g\mu_B B M$. The sign of G_{Δ} depends on the e/f and transverse-axis phase convention; its magnitude and the doublet difference are invariant.

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